

ActInf Livestream #027.1 ~ "Active Inference: Applicability to Different Typ

Livestream | 1:51:51

all right

welcome everybody to the active

inference lab today it's active lab live

stream number 27.1 on august 24th 2021

welcome to the active inference lab we

are a participatory online lab that is

communicating learning and practicing

applied active inference you can find us

at the links here on this slide

this is recorded in an archived live

stream so please provide us with

feedback so that we can improve on our

work

all backgrounds and perspectives are

welcome here and we'll follow good video

etiquette for live streams so people can

just raise their hand in jitsi

if they want to speak at any time or

write a question in the live chat

this short link has a calendar of

upcoming live streams and today we're

here in 27.1 on the 24th

and in august 31st we'll follow up with

another discussion on this same paper

and then we haven't yet decided what to

read in the upcoming weeks so if you

want to come on and discuss

that would be cool

today in

active stream 27.1 our goal is to learn

and discuss this very cool paper active

inference applicability to different

types of social organization explained

through reference to industrial

engineering and quality management by

stephen fox

and uh we're gonna go anywhere we want
with the paper or questions and
we're really appreciative to have steven
here so that will be a cool discussion
let's uh

start with just introductions the
non-authors can say
what was interesting to them or what
kind of brought them to this discussion
and then it will be awesome to hear some
context on the paper that we can take
the discussion anywhere so i'm daniel
i'm a postdoc in california

and i'll pass it to dean
yeah good morning my name is dean i'm
i'm in calgary and uh

thing i found really interesting is that
this was the fastest turnaround time
between the 0.0 which is a bit of a
deconstruction

and now this point one which
is now going to be sort of put back
together or reconstructed today
and i just like the
the idea that the sort of the feedback
loop is really tight

so i'm really kind of
both curious and uh
happy that that the feedback uh
is such a quick turnaround and i'll pass
it over to stephen sillett

oh thank you dean yeah um yeah i'm just
really intrigued i'm very excited to
hear more

questions about
how active inference fits in at the
larger social and applied scales and i

know a lot of us here it's kind of a bit
of our thing

is how does this get applied in
different contexts so uh yeah thanks
awesome agreed about the whiplash dot
zero

so stephen welcome please introduce
yourself and then maybe give us some
context

hello everybody so my name's stephen fox
i'm a principal scientist
at the technical research center of
finland

and my background is i left school when
i was 16 i was an apprentice
and
did that

my craftsman master craftsman did manual
work for 10 years

and then when i was working full time in
industry i got a phd in engineering
design

and subsequently i've got a doctor of
science degree in production engineering
and i came across
um

active inference

and i thought it looked pretty
straightforward and useful

but then i started reading some
scientific papers about it

and it became more and more difficult to
understand

at least for me

so i thought the only way to resolve
this was if i

wrote a paper about it and sent it off
to peer review

so three iterations and five thousand
words of critique later
it was published in
the paper that you've read
okay awesome i have a few thoughts on
that and then anyone can raise their
hand
so
one piece was that getting more
difficult to understand active inference
as you went along what
dimension was that you know like what
made you feel that way and is that
different than other areas that you
learned about
um well the scientific papers
about
on the topic they are
they're addressing particular audiences
aren't they
and so for example some papers
are
addressing computer scientists
who are using
established
machine learning
paradigms
so the paper is directed to them and i
guess the papers are sort of
responses to criticisms
that come from that audience
so if a reader's not a member of that
audience
it um
it's not clear
what it's about so i think the core idea
is uh straightforward enough
but

so it's sort of like this the
like the relationship
to
between
thermodynamic
entropy
and information entropy
the the so this is beyond the act of
inference literature the basic idea is
simply enough but of course
if you read a scientific paper about it
the scientific paper about it is about
some
particular issue there
when it's supplied in a very particular
um
for a very particular purpose
so
i think i understand that the
relationship between thermodynamic
entropy and information entropy but if i
read any scientific paper about it
i at least can become
baffled
because it's a very focused issue and of
course in in scientific literature you
need to go beyond the state of the art
so i think that's the thing that
and i asked
a colleague of mine who uh
to to look into the literature as well
and he um
[Music]
he was
baffled enough to decline in my
invitation
complete with a funding possibility
to carry on looking into it so

[Music]

dean thanks for sharing that stephen so
dean go for it and then anyone else with
a question

yeah stephen i was just curious so
you you're kind of alluding to
who's in the room that that you know
who's the audience thing and i'm curious
because you
you

[Music]

sort of understand yourself as being a
bit of a craftsman and a bit of an
artisan so
do you think the paper has
a ready audience or do you think your
your paper can act as as an attractor
state because we're talking about active
inference
um my own my own paper that you've been
reading yeah um well well i think um
well
the first thing is
now it's clear to me
so uh
and also it turned out i didn't know
until it was published that the academic
editor's professor fristen
so at least it's clear to me
what it is
in what
i'm doing
and so i can apply it so for example
i've put it into a proposal
that's currently under review and
hopefully get some funding
and then
if it's

of
use to others then that's good
would be uh
and i i've had one person who's uh
read it
very thoroughly and said it was he said
it was a great help to him
and he wished he
this paper had existed
three years ago because it would have
saved him so much trouble of
going on the treasure hunt through the
literature
so that's uh that gentleman is in in the
world of business
so he also had been on this treasure
hunt through the literature
finding a sentence here and a sentence
there that he could
sort of
put together and
think well do i understand this don't i
understand it
so yeah hopefully it's of help to others
but it's not i mean i don't think
there's a big there's not a big audience
for this i don't think
it's a niche
it'll always be
a niche
um and it makes me think about how the
development of this niche and the
community dynamics within it have been
different than maybe other fields like
what you said again about getting more
difficult or perceiving the the
implications or the difficulty to be
higher than when you initially jumped in

first off is a is a meta learning
characteristic you're updating your
uncertainty or your scope or the
policies that you're going to have to
take as you explore the space
and then there's um something about this
field that
impels a lot of people to
learn theory and to explore applications
whereas
i don't know other related fields
sometimes like you said they're either
framed as a technical debate like people
have shown this relationship under this
condition here we show up for this other
condition and then that's a bit too
abstracted away from uh meaning for many
people or it's just in some corner of
the literature
um or just sort of it remains at a
qualitative
sort of okay cool cybernetics like goal
orientation and then it can veer off and
maybe get disconnected from some of the
formalisms
and so you mentioned the core of active
inference or that now you have a clearer
sense of it so what did you initially or
what is your more refined sense of what
that is now
um
so
well
illumination came to me when i read
a paper
i think it's an uncertainty and stress
and it says um a professor fristen's one
of the the authors

and it says that free energy
something like this the free energy
is the information that somebody is
lacking
to make their internal model
match
reality
and of course we know then we could be
discussing
forever and a day about what reality is
but
i thought that was a so on the treasure
hunt
that's uh a sparkling jewel of
information because i thought okay so ah
so it's an information gap
it's an info what we're talking about is
the information gap
between the internal generative model
and
the external state
and
what the uh and this variational free
energy because of course our
internal state keeps changing
and the external state of the world
keeps changing
whether we want it to or not
so
as we try to
uh and as actions haven't the actions
we're gonna take we haven't taken them
yet
so
we seek to reduce expected free energy
over a series of actions
and so to me what what is that mean it
means we we seek

to reduce the information gap
between
our internal generative model of the
world
and the world um
which we are in
that we're in
um
so
and then active inferences well how do
we do that we can the actions we can
take we can either
we can change what we pay attention to
that's an action
or we can take action in the world
to change uh the sensory inputs
so that's it
so
and if
so if
[Music]
now this
reducing the information gap
i'm as you'll notice in the paper i'm
using this term unwanted surprise
so surprise all is the difference
between
what we are predicting and then the
sensory input and
the world itself
so
these things can be
for example
in
consumer culture
survival may mean
selecting the right
garb and accessories so one can survive

in a social group and
prosper in there maintain one social
capital so if we're in the if we're in
the rich world
survival can be at that level
but if we're in the uh
other parts of the world survival can be
the difference between literally between
energy output and energy input
over the course of a week and you
literally might not survive if you don't
get that right
so does that make sense
very interesting description
just to kind of trace out the um order
that you brought up a few
the variance of free energy in so you
you
framed it as being free energy is the
information gap
between the internal generative model
and then the external states dynamics so
that makes me think about like trading
like if you knew
what the shape of the
stock or the crypto is going to look
like you could trade perfectly and then
for other situations it's also like if
you knew perfectly what was going to
happen you could act perfectly you could
steer your ship so that given the exact
wind distribution you are always like
navigating at the fastest possible
uh rate
but that is kind of an intractable
calculation to make
due to for other reasons um the internal
and external states changing

and that's leading us to the variational
free energy
and then that is a statistical
calculation conditioned on action which
leads us to the expected free energy
which is seeking to minimize free energy
indirectly through
expected free energy directly
and then you pointed out that there was
this capacity to change attention
and to change
sensory states through action so i think
that was a very succinct uh
run-through of a lot of the active
inference things dinner stephen what did
that make you think about
well when i'm done
could i just could i just read could i
just read exactly what it says in the
paper yeah uncertain to interest it says
free energy
can be regarded as the information a
person is lacking
and which uh
he strokes she could use
to make
his her internal model
as close
as to reality
and the authors of this paper
uh
then these are
so there's professor frison but the
other authors are being doing resurgence
right
so there's a bruce mcewen and carl
fristen
it's called stressing

stress and uncertainty okay why it
causes diseases and how it is mastered
by the brain
and it's published in
progress in neurobiology got it thank
you
cool
so that was the paper that kind of gave
you the insight into the of
free energy
yes where where's the chat here i can
send you the link in the chat
um it's on the bottom left or you can
email it afterwards and we'll add it to
the youtube uh description
so cool
uh
dean what were you um
uh gonna say um on that
no i'm just gonna ask i was just gonna
ask uh stephen
so when i was going through the paper
until i got
i don't know probably 80 in and started
seeing the references that you made to
the to the wayfinding aspect in the in
the exploration piece prior to that
there was a lot of
there was a lot of attention given over
to the precision piece
around the free energy principle and
what i was
one of the things that i commented yet
in the two-point 27-0 was
if we're talking about
active inference we're we're
we're basically talking about two things
at once we're talking about precision

and we're also talking
about provisioning and i think my
my simple metaphor is are we talking
about a hand that fits into a glove or
are we talking about the provisioning
piece which is
how many different gloves
have we provisioned in terms of the
context that the hand might find itself
in
cold weather water
forging a piece of hot iron or whatever
and and so what i was
what i was curious about was in terms of
the social organization in the
industrial engineering piece
how the
how the provisioning and the precision
coincide with one another in terms of
what you
are thinking now post hoc writing this
paper
can i
can i share my screen somehow yes
let me um unshare these slides so on the
bottom left the middle button
share your screen and then i'll make it
um like full screen so you can make it
look however you want on your site
[Music]
sure
all right
okay can you see him yeah hit the hide
on jitsi where it says
uh hide the sharing your screen
and then um yeah i can make that full
screen so thanks
so you see a diagram yep figure one

okay so

why there's so much

um

emphasis

on what's going on in uh

down here in make to stock and assemble

towards

is because

what i was trying to do was get a

an anchoring

so

okay in

here

in where where there's um

mass production and mass customization

where there's a great deal of refinement

of what is offered through many rapid

iterations

but then there's something

that's fixed that

okay there are there are a lot of

activities like this in the world and

the diagram

about the quality

process

so this is in in a way this is as if we

think about it every evolution

like the toyota corolla

is a highly evolved species

compared to um

if we make uh

if we make a vehicle ourselves from

scrap

so i don't know it's been in the news

there but a teenage teenager in india

has made a tuk-tuk out of scrap

solar powered as well

so then

that's a

it's not highly evolved it's the first

prototype working prototype

and then these things up here this uh

top engineer to order

the initial situated entropy

is

much higher and the barriers to

information gain are much higher

so

as the diagram you're looking at just

changed

yeah

okay

so this is from another so that that

diagram i just showed a minute ago

that's from another one of my papers in

entropy and this is from another one

so what's going on is is that um

if so with uh

diy

dude

do i do it with others presumption

uh massive engineering

but we it begins with a just an idea in

our

in our minds

and we may

begin at the level of uh i think this in

the video did you show a carved piece

so it might be you've got a

a tree stump in your back garden or you

might have a tree and start down there

raw materials and you might

shape it and then there can be this

phenomena that well

there's a dialoguing between

the human doing the shaping

and the piece of wood itself because the
piece of wood has got its own properties
and
you're not going to get a
an aesthetically pleasing carving if you
chisel against the grain right so so the
grain that has been shaped
through perhaps hundreds of years some
of the trees there in california
um
so there's a dialoguing
between the
the human
doing the shaping
and the the piece of wood itself
shaping what the human does
but then more
from an industrial point of view
uh if um
anything
that is bespoke
so the term apparel is more inner
the term bespoke is more appropriate in
apparel than um
engineer to order so bespoke is short
for bespoken for
and it meant be spoken for by an
individual
so the materials that were to be used
would be spoken for by an individual
and in today
what happens with bespoke is that
there'd be cloth
so that's a formed material
parts buttons
and
then
suppose it's show business or

suppose that elton john in his most
flamboyant days had an idea in his mind
for a new uh costume
it would be a preliminary idea
there'd be iterations of sketching
and uh doing some mock-ups maybe and
looking at the fabrics and they'd do it
and then maybe we get to quite an
advanced stage of the fitting and he'd
decide that
he'd look at it and think i don't like
this actually and completely change his
mind and they do something else
and then with engineer to order
that could be for example uh the late
steve jobs is yacht
so he might have an idea in his mind
about uh what it might be what it might
look like
and then the the ship builders
they're going to try and get as far up
the bill of materials as they can
so they're going to try to use the um
the cabin modules
that they've used on other ships
or at the very least to use the uh
the 3d digital models that they've got
so
what's in the paper
it's really
around this um
it's it's down here
because then
that's the
so rather so to start with something
that's quite a concrete
and highly transferable
before

then as i do later in the manuscript in
the paper

um

getting into the project

processes

and

the way finding and so on

does that answer the question

can i just do a quick follow-up just so

it's clear in my mind um

so what i'm seeing here what you put up

on the screen is

and i don't i don't i want to be

corrected if i if i haven't got this

right is i see that i see

ways of laying out a a fabrication

series

and i guess what i was wondering is more

on a philosophical level

as you're transferring from idea to

fabrication

[Music]

how do we use active inference there

how do we

turn the idea

and translate it into this

fabrication exercise is there

is it a is it

is it a planning effort is it a

is it a

getting our ideas

plural provisioned is it about

propositioning

the the garment maker before they

they go on to to do the construction is

it

you know i i i what i'm trying to do is

get in my head clear

what your focus was on because i could i
can tell now already
40 minutes into our conversation today
that
there was much more you could have
written about but i'm i'm sure basically
what you wanted to do was
was
get one
one thing clear in terms of this paper
but that for for people like me
that opens up a whole pandora's box so i
just want to be in my mind
certain that
the focus here is about the fabrication
piece
and not the stuff that comes before the
fabrication piece of my
am i leaning in the right direction
seems like he silently left actually
oh no sorry oh no
i was on mute oh continue then
yes
um
yeah so in the in the paper itself
um i think i'm using the term
it's mass production and project
production is it yeah
so
in
so yes other
the focus
on the paper of the paper is on this
mass production
to just to try to
what's the word i'm looking for i can't
think of anything better but just just
just to

sort that out just say okay

so

that's what happens in mass production
and here are these diagrams and these
explanations here and there's the the um
statistical process control charts and
things like that

that's what goes on there that's settled
now

we move on

to these project the project production
and why i'm focusing on physical
production is because

it's then

it's tied into um

[Music]

action

it's the there's a clear link between
action and energy
and of course energies in this is a
finite supply of it

uh for individuals and organizations

so

the mass production stuff

is to provide the

the platform

to go into more challenging areas

am i answering your question dean

yeah

tangential rambling

no no no no

like

it i guess it depends on how much you
desire decide to zoom out

um and so i i just want to be clear in
my head

what the focus is on because there is
absolutely active inference in

fabrication and manufacturing i have no
no quarrel with that whatsoever i just
want to be able to make sure that i
don't
overextend what i think is the act of
inference
potential
because as ever said when i was reading
the paper
where it got exciting for me was that
whole exploration
beyond the precision and on to the
provisioning or provisionings part so
thank you yes you
you cleared that up for me
awesome
uh
how about stephen with ray's hand and
then anyone else
yeah thank you um
i i'm i'm looking at this here and i see
some interesting um ways that you're
engaging
free energy in relation to
active inference because often we we
might see mathematical models
and then philosophical arguments and i
sense that this mid-round which may be
where a lot of the applied work happens
at scale is is heuristic
based
or principles based to sort of get a
sense of how to navigate
um
and may not be directly modelable
because there's so many dimensions so
that's kind of interesting so i wonder
you know where where people are talking

there about situated entropy or

the the the

um

it's it's a potential

um

so i'm wondering how you how you think

about the heuristic and how you think

about

um

how much someone could model this type

of thing or how much it's about

modelling and if so what type of

modeling

might be useful

in organizational industry contexts

um

well i think at the end of the the paper

i i raised this question i think about

um

well

there's uh

there's maths

and then there's modeling

and then there's uh

simple rules

and so in engineering design

uh

and not just engineering design uh

simple rules can be extremely

so designed for manufacturing assembly

the application of that has led to

90 reductions

in the assembly times of many products

so

you could

you could have just

kept modeling the

the former process

as people did
for
for many years with ever more
sophisticated
mathematical models and simulation tools
but it still
still wasn't very good
or you could just apply some simple
rules reduce assembly time by 90 percent
and then
well then it's a lot better isn't it so
i'm
i sat um in one of the uh
the meetings that um
professor fristen and
maxwell
ramstad i think we're
going through professors fristen's book
[Music]
the monograph the name of which escapes
me now a free energy principle for a
particular physics
yes
so
professor fristen and thomas parr were
in
discussion with
is it michael
bail who's
uh he works for a japanese ai firm uh
martin deal
yes money bill yeah
and um
at least one colleague of
martins
and they were discussing amongst
themselves
and

i don't think as far as i could tell
they weren't able to get
an intercepted
understanding between themselves even
and martin's published a paper
in entropy
where he's contesting some of the maths
and
so
my view is that
of course things develop and they have
to start somewhere and
no doubt that will lead to good things
but it would be more people that can
understand something the more likely it
is
that
opportunities for its development
and flaws in its current um
framing
can be identified
and progressed
so
i'm and i've sent him my email this
morning this quote from einstein didn't
know
from
from geometry and experience that
uh
what do you have it there now um
look at that quote from al einstein
it was in my email um i can look it up
in just one second but right now just on
the screen was the um guest stream
with martin
and so
totally agreed really fascinating there
um okay

so
answer the question i'm
i see of course there's
there's a place for maths there's a
place for computer simulation
but if so if we go for
so it may be
that some are engaged in positivistic
research
and they're they're seeking to establish
a correspondence truth
and others
you know anti-positivists interpretives
they are
engaged in
looking for constructivist truth
and what i'm doing is i suppose
it's critical realist pragmatic truth
so if it works it's true
so
what i'm doing is well
it's probably a bit rough
but maybe it's good enough to get
something done
yep very
interesting um
one
question i had was
you talked about that kind of
co-evolution with the iterations whether
it's the the craftsperson working with a
raw material with its imperfections or
its characteristics
and then all the way up to this more
social form which is like uh elton john
and the costume designers in in that
feedback so whether it's just
hands on object like drawing or whether

it's it's more extended to include all
these other aspects
how do we
hold space for that versioning and
co-evolving process
and for the material to allow it to
reveal its characteristics while also
having effective supply chains
you know that can make cpus and that
deliver
high precision materials
because it just sounded like you had
experience on on both sides with the
industrial engineering as well as the
craft and artists inside
yes so um
well
with uh
automated engineering tools yeah as you
i'm sure you know there are these
communities um
of makers
and uh
where people are sharing
is instructable still going
as a
online community of giving instructions
for projects and things so
so one
one big
challenge
is that it's skill so manual skill
so if
if
work like carving
it depends upon um
quite a lot of manual skill to get a
reasonable looking carving that

somebody might be happy with

um

but if

you're going to make something if you're

going to 3d print something

then

the

the manual skill required isn't so

so high and that can be communicated

so some of the uh work that i've been

doing is in this virtual social physical

convergence

but

[Music]

through virtual technology

you can share socially skill knowledge

in doing physical work

and

if also um

do you know the open ecology

movement where there are engineering

designs uh freely available

on the internet

so through virtual social physical

convergence

there's great potential to enabling more

people

to

produce what they consume to become

prosumers

and as gandhi i paraphrase gandhi now he

said prosperity can't come from mass

production

it can only come by from production by

the masses

interesting answer

reminds me a lot of bucky fuller trying

to pivot the supply chain from weaponry

to livingry mass producing the aluminum
sheets for housing but that was
kind of
only potentially a transient step
in this supply chain in production and
then eventually knowledge reimagining
will we all have a silicon foundry in
our house
probably not
could we all have
a compost bin or could we all have
a sewing kit
that's something that's a totally
different type of
artifact so it can be
mapped differently
dean and then anyone else or any
questions in live chat
well i don't want to go off into into
the woods here but i was curious uh
stephen if
if you've ever read any of this stuff
from
matthew crawford
in particular
um
shopkraft title and then the other one
is
the world beyond your head and the
reason why i'm asking is because in that
in that second book which is one of my
favorites he
he talks about all of the distractions
in the world that cause us to not be
able to be focused
and precise and do
the the crafting never mind being able
to do the

the uh
indus you know projection at an
industrial scale
and so i just i was just curious if
if maybe there was um
maybe that author was maybe whispering
in your ear when you were putting this
together because
if you whether you're whether you're
carving something out of a piece of wood
and having a dialogue with it which i
completely agree with or
you're just making sure that there's no
bugs in the
in the command software for the for the
making of the copies
it's still an ability to
be able to use things like active he
wrote he wrote the beyond your head book
in 2014 so i don't think he was
looking at active inference but i think
that
how that is employed
can sometimes be
perceived as
just the right amount of being able to
give attention over to something and
still remain open to
possibility so i was just wondering if
that was something
that may have been uh
adding some color to what you were doing
here
no i had not heard of that before
but i shall i shall look into it with
interest
yeah it's fantastic stuff
yeah thanks for the comment dean

stephen

yeah this um

i hope this does again doesn't go in the weeds either but

but i think this is like say this this touches on so many things is when you talk about the generative model

um

and you've got in body cognition sort of fitting into that um

generative models

often be thinking that it's an action orientated model so it's an act it's a model of how to act

okay

um

and i'm curious how far for instance if you do welding you mentioned welding and stuff like that

so

with this skilled performance idea that can be

the generative model of how to weld but at what point or this sort of maybe ties into something i've been exploring at what point do you think that the instructions to weld

are in the niche

and being used by someone's generative model of which they only have a certain sequence available you know

um now welding it may be that someone just they get a skillful practice and go the whole way you know so it's all in there but i'm just wondering once you get to sort of this management level at what point it's

a niche

artifact as much as it is a generative
model internally

um

well i think uh

with craft skills

it's

being able to do it it's just at some
point or other after as well as an
apprentice i think most people are
apprentices have had this feeling that
you're struggling and then
you wake up one morning and then
suddenly you've you've got a feel for it
you can do it

and um

exactly

how that is

um

[Music]

i don't think anybody really knows
because somehow it's going into the
the fascia system that's the thing
the feel for it is embodied in the
fascia system
and most
um
to my reading as a craftsperson
most literature about embodied cognition
is just paying
lip service to it being about embodied
cognition
and it's not really
addressing the nature of embodied
cognition about
how much that depends upon the fascia
system
and body memory
and

in this 4e model
about extended cognition
you know put put your memories in your
mobile phone type of thing
but there's no way that
you can put your
your manual skill
uh embodied cognition
muscle memory
in a mobile phone you can have many
videos of
people doing manual work from all kinds
of different angles
but
so that can be informative
but it's not
uh as as you can have many videos of
people riding bicycles
that
that's not the same as
having your body memory of how to get on
a bike and not fall off while you're
pedaling it
um yeah stephen go further than i ever
thought
yeah i think this is a good i i agree i
mean there's so there's some transition
that happens when
it goes from presumably some
instructional process which is being
decoded in a in a kind of a
cognitive way to something more embodied
like you say and it becomes a practice
but then there's also i suppose and i
don't know if but when you start
coordinating for instance you've got a
whole group of welders working on the
side

of a
oil tanker
so there's there's kind of then it might
start to move again
there's a scale beyond the individual
welder you know and at what point does
it then
get beyond what i've been thinking of as
like an experiential
ecosystemic kind of
knowing
something which has to be coordinated
through some artifact like a plan or
something
cool
um let's just go to blue first
go ahead because your thought is
probably my thought oh no
you should go for it
so i have a question that was kind of
related to steven's question um in terms
of
the niche modification um as not as it
relates to craftsmanship per se but to
industrial engineering and and in the
this paper that um was the topic of our
discussion
you said that um you know the
the processes need to be carried out
right the first time in like 99 of the
cases so i remember this and and to what
extent is that first time right i mean i
i just remember as a business owner
there's a lot of sop development right
so you develop sops and kpis so that
everybody knows exactly what their job
is exactly what they're supposed to do
and exactly the procedure for

executing the job so that theoretically
anyone can come in and be like subsumed
under the markov blanket
for the business and the niche has been
modified maybe in such a way to make
that happen correctly 99 of the time the
first time
so do you think that that process
development in in the industrial
engineering application is
niche modification or perhaps something
else
um yeah so this the 99
times out of a hundred that's so that's
six sigma
which is
under the umbrella of total quality
management
and
that is where
uh so if you if you've made
so this is all by the way are you still
seeing my screen
uh no uh maybe could you reshare it
okay yeah hang on so
let me just
if i go back to um
[Music]
so if i go
back to that point i was making about
some embodied cognition
so
yep looks good this
this can you see that certain amount of
predictive processing yep
okay
so
what i'm doing in this paper is i'm uh

refer i'm making reference
to the fascia system and body memory
um
and relating
you know well predictive global neural
weight workspace
um
so
that's um as far as i've got with that
now this 99
now can you see the treadmill yeah
okay there it is
this this 99 uh
at a time right this time this has come
with the treadmill of production and
consumption
so if if an organization invests a
billion dollars
in building a factory and uh
and then the tooling
then
it's got to mass produce and it's got to
get people to
to buy what it's selling
and they can't be disgruntled customers
because now there's global competition
and
that 99
it can be measured to microns with car
body panels and things like that
but if if you're in uh the leather
um
the leather
cluster in italy
and you're making leather bags as has
been going on for hundreds of years
then
the precision

is not so desirable
it's good
if every one of the leather bags is made
that is made
is a little bit different
to each other
because that
signals
that it's handmade
so the 99
right first time
he's all wrapped up
in this um
treadmill of production and consumption
with varying levels of precision
depending on what the niche is so if the
niche is
you're a global car maker
well
the door better close with the
obligatory clunk
and it better be a water tight
airtight
fit
when you're driving at maximum speed
otherwise customers will be lost
so
in different niches
the level of vision that's required
depends upon what customers
are expecting
and what the competition will tolerate
did that did i answer the question at
all
thanks for the
interesting answer i'll all give a
thought and then we'll go to scott so
first um it's almost like what about a

semantic precision like a writer who is
outputting um precision in terms at
multiple levels like a typographical
precision and also um like they don't
just want to release you know their hand
hitting the keyboard that's not what
their writing product is so it's not
just the mechanical precision that we
can talk about and then that idea of
waking up to skill and the way you
connected it to embodiment is really
interesting maybe people wouldn't expect
that from an industrial designer but
then when we think about craft in the
continuum it makes more sense and then
also that waking up aspect of it it's
like during sleep when we're simplifying
and pruning neural connections
the more neural
applications of free energy principle in
active inference could help us see that
as like a process of model
simplification and unlearning the
details and that is sort of where skill
and mastery
uh comes in and that could be cognitive
unlearnings and bodily
updatings in terms of the morphogenesis
of the tendons or
the sensory receptors or
all kinds of other
changes and so it's just cool that we
can like really
telescope in on just wanting to apply it
to as you did in the paper really
partitioning out the biological systems
partitioning out the flesh and bone
but we know we can go there

but there's there's so much that you can
say about just the engineering side
that it was just like it makes it very
interpretable yet also clearly
comprehensive which is what you pointed
out was kind of lacking in the
literature when you were learning
so scott um go for the question i don't
know if you watched the dot zero but we
we we uh
we we had a scott david alert with the
markov blankets of contracts and
but yeah anyway just go for it
yeah thank you sorry i was delayed i was
on another call a couple of things that
thank you a couple of interesting
comments that um
sparked some questions ahead so the
talking about the fascia and the and
your statement about the mobile phone
can't record the fashion obviously
that's true but it sparked three things
to me i wanted to get your thoughts on
one is
what in the notion of the fascia and the
embodiment
is it possible to use active inference
framings or is it fruitful in the
opposition of various fascia elements so
when you're learning diving or frisbee
playing one muscle is pushing against a
bone or whatever and so the fascia has
multiple components as well
and so are there learnings that happen
among fascia elements themselves
because they're building actin myosin
you know
bulk or whatever i guess those are the

proteins but in any event they're building bulk that's one is the opposition among fashion elements the other one is it reminded me a little bit on the mobile phone thing sandy pentland at mit talks about i think he did some work on predicting if people are going to get alzheimer's by they have their mobile phone in their pocket and they start walking around their apartment differently and the mobile phone is able to track that there's a difference an anomaly and so i thought that was kind of interesting that although the mobile phone can't track your craft your but it can note anomalies based on picking up fascia

histories and then the third thing it made me think a little bit about um robotic prosthesis and haptic sensors and so there again you have this feedback mechanisms that are bounded obviously again they're not storing craft but it seems like i just wondered if any of those sparked any um thoughts that

relative that question of the i guess not just the mobile phone but that sensory apparatus and the information networks being responsive with elements of fascia in different direct and indirect ways and maybe an active inference framing a hybrid between the technology embodying a certain expectation let's say and the fascia embodying a certain expectation

and that kind of
harmonic coupling if that had any any
thoughts on that thank you
um i think
one thing that i'd consider in all these
things is
uh
what's the
but
i'm going to mention the r word now
reward
uh what's the uh
what would be the effort to reward ratio
of doing that so if
if um like you said about learning to
dive
so i'd say that maybe there would be
some uh
rules
um
sort of like design rules for learning
to dive that maybe could be
formulated
that might be useful for people to
learning to dive
but then going into
very sophisticated
modeling when individuals are so
different
that might require a huge amount of
effort and the reward
may be less so i don't think you're
alone when i was saying before scott
that uh
that in
industrial engineering
um
so up until 1980 people use the

mathematical models and the simulation

tools available

to to to model uh industrial processes

but

then

designed for manufacturing assembly was

introduced and assembly times in many

products was reduced by 90 percent

with simple rules

and so all the amount of there's

there's a place for modeling but

what i'm more interested in phase

focusing on

are there

simple rules

are there

simple methodologies

tables and diagrams

that

can provide insights as designed for

manufacturing assembly did that can lead

to very substantial improvements

so that's

that's what i think but

but then like this gate tracking you

were talking about about people are they

are they getting out on is that

if

one thing that i think's very attractive

about active inferences it can provide a

starting point

where

humans

and artificial intelligence can be

starting from the same place

that okay

what what's going on here

is we're trying to

reduce this information gap
between
the internal generative model of the
human being
and
the external state of the world
that they're moving around in
and
[Music]
both the human being trying to do the
under and the artificial intelligence is
trying to do that
and how is that done it's done through
active inference
where
we learn
from
either
changing what we pay attention to and or
changing the action we take
so then there can be a common starting
point and maybe this can make the human
being feel
much more comfortable
around the artificial intelligence
so i've got i'm working on a
project at the moment that's this third
wave of ai where there's some
top down rules
and
bottom up sensory inputs
combining
and
i see active inferences being
well worth exploring as a starting point
for
human
ai interaction

where both the human and the ai have got
these well this this generative model
that's the humans
it's it's it's understandable to the
human
awesome points about
where active inference could come into
play and also almost from a designer's
perspective if they have a systems
diagram
uh or a digital twin of some type that
includes
active inference agents interacting
first because they're all of a similar
type of model they'll be increased
interoperability and ease of computing
these kinds of systems
and then
the system designer or perturber can
simulating with counterfactuals or with
any other technique can
uh formalize what affordances they do
and don't have so it's like if someone
says the only way i'm going to get this
marketing um you know to succeed is if i
update their generative models so that
they think that you know this isn't a
ponzi scheme it's like if that can be
formalized then we could have benevolent
shapings of the niche
and kinds of interjections like we're
going to inject this much money with
this much of an equation
because we're doing kl optimal control
on this parameter conditioned on these
other parameters you know how do people
feel about that versus
we're gonna do an ad hoc strategy based

upon conversations that you weren't part
of
so
and then that just while you were saying
about the generalized principles
again really interesting because there
was a lot of detail in the paper
and a lot of
really insightful comments but it wasn't
um the same formalism-driven
approach
that we've seen in some other works
directly of course cited a lot of the
primary literature there
so it's just cool how you pulled back to
the level of the generalized principles
and then we can dip down into more
special mathematical framings
but this last part of table 1.
which were also all sections of the
paper or like essential kind of
principles
that
at least
you and the reviewers agreed were
important enough to define
clearly
yeah i found that that quote now so
this is from
[Music]
albert einstein geometry and experience
addressed to the prussian academy of
sciences in berlin
27th of january 1921
as far as the law of mathematics refer
to reality
they are not certain
and as far as they are certain they do

not refer to reality
nice that reminds me also the way that
you again describe free energy as that
information gap
it's almost like we can have a
description
or an estimation of of a gap a negative
space
and so it's like well what is it what's
what's in the gap it's like well if i
knew
or if we had any more precision then
we'd tell you but all we know is how
large
the deviation is
uh so it's sort of like a way of
uh one way around it is like using a
positive definition of something that's
lacking and then that's closely related
to this idea of tractable minimization
that is better suited to be calculated
than an intractable maximization
like maximization of reward or expected
utility which is a totally different set
of constraints and outcomes than
a
variational free energy gradient descent
so scott and then anyone else
i wanted to you know that got me
thinking are they just the
uh i guess the
um library of opportunities for a system
or i'm not sure how to characterize it
but um one of the things i've been
fussing with is the chris christopher
alexander um for folks who haven't
been in that space
with i just learned that a few months

ago about him but
he's an architect who collected the idea
um that he had had was the collector uh
he was in in architecture and is
collecting up different practices in
architecture and different patterns
and he came up with a pattern language
the idea was if you want to create a
portico and a
place to negotiate
private and public space for instance
you had certain patterns that happen in
different kinds of architecture and so
if you wanted to do architecture you
could look at a bunch of patterns and
and stare at them for a while and see
which one look best
and it's interesting with these
questions of filling gaps you know the
easiest place to look is what other
similarly situated parties have done in
similar situations so if you have that
crosstalk the peer-to-peer conversation
um then you can discover without having
to go through all the pain and the cost
of discovering yourself and that's the
notion of guilds right i heard reference
earlier to apprenticeship
learned by the masters not just
reinventing the wheel kind of idea okay
so
and it's interesting when we talk about
gap filling because in
more static systems culturally static
systems let's say historically quiescent
syst periods
then you can learn a lot from the guilds
but in periods of high dynamic change

environmental change
the guilds are going to maybe under
serve you because there may be so much
change that the old practices just don't
hold up the externalities are such they
swamp things
and so i wonder in that gap if there's
an element of discovery
both of the
available what what are the ways to
discover what's available
and then discover the things that are
not available a because they're in a
different silo so they haven't been
available but maybe because they haven't
yet been tried in any silo right so
they're just not they're emerging
type of notions and maybe that's a bad
question i don't know it just it felt
like that gap
there's there's an easy low-hanging
fruit of practices chris alexander kind
of stuff is relatively low hanging and
then it seems like there's other
mechanisms here of discovery
and maybe that's randomness and maybe
that has to do with evolution but anyway
if that's commentable i'd like some
comments on that yeah awesome question
scott
um are you are you still seeing my
screen yes i'll make it big
all right okay so now
skills
leapfrog skills yes there it is
so um
the thing is
about manual skills

as scott was just saying the guilds and
everything so
in the work that i do development
projects that
point out that
industrial thinking is 300 years old it
came from the english midlands 300 years
ago and
and taking that as the uh there's the
basis of um
doing development work may not be a good
idea because it's 300 years old and the
world's change
and so it is with skills
that
um
it's good to be able to carve wooden
pieces it can be very rewarding
and maybe it can bring a few people some
economic benefits but that can't uh
address the poverty of billions of
people
and
craft guilds
as i
say to people well when they say
ministers when they say well we need to
build training centers and give people
apprenticeships and i point out well
that's medieval thinking
so
that
isn't good that's not very good
so here there's explanation of leapfrog
skills i'd take you down to
some diagrams here
oops
just

a simple example
bear with me
this is taking longer than were you were
you what was your active inference
uh stage or
awareness at this time in 2016 or when
you're writing it leading up to here
zero
i didn't
well no but the thing is it looks it
depends how you look at it doesn't it
because
i
so if you're doing manual work then
you're doing active inference
because
it what what oh what are we doing this
uh right so information gap how am i
going to do this
because i need to do this because if i
don't do this i'm not going to get paid
if i don't get paid
i can't buy food i can't pay the rent so
how am i going to do that that's the
information gap
and then so i'm going to pay attention
to
my
priors
what have i done before
i'm going to
pay attention to
this the sensory inputs from the
external state and notice that
oh
that the not all the materials i need
are here
so i need to improvise

adaptive expertise i need to
not apply
what i've done before
where it's not gonna help and i'm gonna
have to think of a bit of something to
do it
and uh so in my opinion i active
inference is certainly
what's going on in manual work
and then you you give it a try
it doesn't uh
all right it works to some extent or not
to another
extent so then look back in the
generative model
hang on what have i done i've done
something like this before look around
oh look i see in the distance there's
some bit of kit i can use over there
i'll use that and so on
um but to go back to this craft uh
thing that
can you see that figure five that yep
the kind of stuff you put around the
door so
doing these joints
the difficult form to fix
and
very often if you stare at such joints
closely you you'll see they'll have been
filled up with glue and dust or
a bit of
more paint being stuck in there
but then if you do this instead
it actually looks better
and
it doesn't no need for the craft gill
guild it's just a kit

it's just um
these molds it's just a kit
so you've got the and these uh
blocks here in the corner
it makes it look more fancy
and you can just assemble it
so if i go back to this one
okay so
do you see now a diagram that says
movable factories yep
okay
so
movable factories on the job training
product kits in the middle
uh engineering design of capable
processes for mistake proof production
though
this is
as we know ikea spread around the world
much more successful than any furniture
company
its competitors because they do
engineering design of capable processes
for mistake proof production
and this can be applied to
many things water towers
solar arrays
anything
really
so this
in uh what scott was saying about the
craft
guilds that
yes craft guilds although i'm a master
craftsman myself
it's medieval thinking
and it's um
it's just

not scalable
and it's very very wasteful hugely
wasteful and really craft skills
are
good
for coping with chaos
but chaos is
hugely wasteful
awash with entropy that can't be
resolved
so in in when i'm doing development
projects this is what i'm
advocating
what's in that diagram
uh dean with a raised hand and then
anyone else who raises their hand
i'm just
throwing down a couple of things here so
it's kind of it's in
this is a nice fit into the glove of
what scott was asking and then what i
saw in terms of the title of that paper
so my question is this if we um
using the same kind of triangular shape
as the five why's
if we start at the top with learning
and that learning then breaking down
into
one aspect of which is updating
and then that breaking down into
one act of which is gap respecting
that then breaking down into assessing
what the step function is
and then that breaking down finally into
some sort of action taken which is the
filling to follow or the gap closing
is that medieval thinking or is that an
invention gradient descent

i know that's a lot of stuff but
if if you want to take the best of what
happened 300 years ago and not get
locked into starting with i'm going to
close a gap because i perceive a gap
present
is there a way that we can take the best
of
the past meld it with the current and
come up with an inventive gradient
descent
which which doesn't sort of hold us
to either the craft or the mass
production
i believe we can
good
we'll be good
friends yes
and in um
in that
paper
let's see if we can get to this
um
there's yep uncertainty and stress
nice questions
[Music]
what you said about the scalability of
the mentorship
in the guild model versus alternative
models was really interesting because
there's the uh sort of academia versus
non-academia
divide so with that sort of student
through phd and that traditional
mentorship system and how that's
evolving then there's the traditional
craft
um trajectory the apprentice and the

master

and the way that that is also changing
and the way that academia and research
are

reconnecting so there's just so many
cool threads about how everybody will
find guidance

so are you are you seeing my screen yep
mass imagineering mass customization
mass production yep

so i'm

i'm looking at

in this one i'm looking at it from the
point of view of consumer culture
and saying how these complement each
other

so

uh

it's in their deen about how these
can all be complementing each other
i'm definitely going to have a read of
that

remember to maintain firm downward
pressure on the soles of your feet while
you do

and pump to the tempo of staying alive
no no no you've got to keep your feet on
the ground otherwise it'll blow your
socks off

scott

yeah i want to get back to um that was
you struck me with that statement before
the craft skills are very wasteful
and they're useful with chaotic
situations but chaos is really wasteful
that's really that struck me and i
wanted to ask you i
thought well that's interesting because

inefficiency is resilient
and distributed makes me think about
distributed agriculture distributed
manufacturing a lot of the resilience
discussions that are going on now i work
with a group called all fed that's
trying to figure out how to feed
everyone in the world and things like
that and
then also the notions of monoculture and
the production of cheap industrial
calories
but then there's no variety
and again getting back to this
the notion in active inference i keep
coming back to carnot's characterization
in the second law of thermodynamics and
you need the hot and cold differentials
to perform work with a heat engine
and i feel like you need the
differentials and information to have
active inference bayesian
occurrence
so
that inefficiency feels like it's laden
with value because of the differentials
that are in there the chaos and the
it feels like there's
lots of value to be derived from chaos
and inefficiency
and maybe that's part of what it was
well i guess it was being done in the
industrial revolution notions as you
took craft skills and reduced the
inefficiencies and derived value and it
feels like as we have more chaos in the
information sphere
are we how might we release those

that value
but but still recognizing that maybe
it's a there's a font of inefficiency
that keeps coming at us because
interaction volumes are going up so we
don't have to worry about
inexhaustible supply of inefficiency
that awaits us but if we mine the
inefficiency and bring it into the
system where it's known then um it seems
like it creates value i wonder is any of
that commentable
i hope that
made some sense
yes yeah yeah
um but before i go on to that also dean
there's this paper
domesticating artificial intelligence
can you is that on the screen yeah
okay so expanding human self expression
through applications of artificial
intelligence in prosumption
so that's another um
another one about
this combining different
different uh paradigms of
production and presumption and
consumption
okay so
yes so what i was saying about um
inefficiency
so
perhaps i should
explain better what i mean
but
redundant variation
so redundant variation doesn't add it
consumes energy

and it doesn't add any value
so that
that is the the thing that with this
leapfrog skills i'm pointing that out
that well it serves no useful purpose to
have redundant variation
variation
can have value but not not redundant
variation so
that's what i mean and also it depends
on what people's circumstances are so
if people are
[Music]
living on the edge
as
many people are
then
there's just no
they don't have any spare
energy
and redundant variation can
uh well we know
do you know this term the prokaryote
that people in in the west that uh
that people don't have they have zero
hours contracts and
and things like that and they're
in the precarious but the real procare
in the world
are the people who were
living on the edge
in developing countries
because
you know they're they're pretty they're
they're staying alive it is literally
precarious
that they they very don't
getting water's difficult and when they

get water it's dirty so they've got a
choice they can drink it
to keep living
but if they drink it
they're gonna get ill
and maybe the illness is gonna kill them
so
there are um
different scenarios and in in
development
work
it's uh redundant variation is just
trapping
can keep people trapped in desperate
poverty
could you just comment a little bit on
as a follow-up on the
expand on that redundant variation what
that looks like in some examples
and also just wanted to mention in one
of the groups i'm working with all fed
we're thinking of developing a how many
days to starvation app
for people and like the countdown clock
and the bulletin of atomic scientists
but how many days a person has in their
food availability including what's in
the refrigerator what's in their
food desert etc but it's days to
starvation because of that awareness
that precariousness but anyway but if
you could talk a little bit more about
redundant variations i'm not quite sure
the concept of what that looks like
is
can you see this diagram again on my
screen this yes
okay

so in that um
in the difficult to form fit and fix
on the left hand side
the shape if there's a little bit of a
twist and um
a little bit of poor machining
in the
timber sections there
it can make forming that joint really
difficult
and then um
[Music]
uh
those so those
variations they're just they're just
redundant they don't they don't add any
value they lead to uh
because there is this this
the duality of entropy
there's a little bit of physical
disorder because maybe the wood's a bit
twisted or the machining was a bit off
so there's the information uncertainty
how am i going to make that joint
so
then
suppose
suppose there's six ways of forming that
joint with equal probability just like a
fair die
and the uh
the information entropy is 2.58 bits
and that 2.58 bits it's just it's just
not adding any value at all
um that gap of 2.58 bit
but with this alternative on the right
hand side
those uh

the square pieces in the corners
they can have a rebate formed in them in
their in their thickness so there's
quite a substantial overlap
there's maybe uh
nine millimeters
or three eighths of an inch overlap so
if if the joint isn't
perfect it just covers it up and it
doesn't matter
so that's a pr a practical example

[Music]

and then if we scale that up to
all the
the many many things that can be so
why uh design management factor and
assembly reduce products assembly times
by 90 is because it just eliminated or
redundant variation
does that help
you know what what that
reminds me of is almost like we talk
about aberrant precision we've talked
about that on the neuro psychiatry side
delusional beliefs um failure modes of
decision making algorithms
it's okay to not be latched on to the
perfect strategy as long as there's this
healthy uncertainty about strategic
decision making because it's that
uncertainty that's getting kind of
coasted on that should the system
persist ends up finding an appropriate
or an adequate strategy but aberrant
precision when there's like high
precision um confidence but then also
it's off target with its mean estimation
things can go awry and so it's like that

door frame on the left the mass produced

one

then

as you've pointed out it's kind of like

a false sense of precision because if

you look at it it looks like the more

precise one it looks like oh it's

sharper or it was machined such and such

angle at exactly this angle but then

that one degree tolerance which sounds

like really accurate on the machining

wheel

now that plays out on the door frame and

now it's actually less precise in

practice

and also like you pointed out the

example with the cupboard or the

bookshelf being attached to the wall

like you can design

for internal precision really well

but then it's hard to take a step back

and have latitude for like the wiggle

room of real life but that's actually

where the success or the failure of the

product in use is going to be

so that's very

interesting topics that you brought up

you know just a quick follow-up my

happiest one of my happiest days in my

life is when i discovered that you can

mix sawdust with epoxy and then do gap

filling and i actually completed

projects a lot faster as a result

one other comment on that is contracts

you know

mass contracts we had a lot of

negotiations i used to work with banks

and software companies and things doing

contracts and it was funny because in negotiations a mass contract has that you get rid of redundant variation right you just get there's no reason to have a different contract for all these things but it constrains the people who have less power in the negotiation but it's funny because people say to you when you're in the negotiation with someone who has a mass contract they'll say well

well you can't change it this is the way we we always have done it and that right they're saying this is just it and then the answer back to that is well let me show you how we're going to do it for the next 20 years you know because it's a world changes around these things that have insufficient variation in them and so it's interesting when what seems like a system that has eliminated redundant variation becomes anachronistic

um
are you seeing my screen though are you seeing three circles yes

okay so scott you were talking about uh distributed

and

so what

i'm

working on is this it's this highly distributed prosumption highly distributed and highly efficient compared to distributed subsistence which is distributed but not efficient and centralized which is efficient but

not distributed

so that's the thing and here is the dark
picture of what happens with
centralization

okay

that looks like the fovea
and retina

the focus all the attention

is on the uh

the heart of the empire

and then it's all dark

so nice images and also that distributed
model is very cool

so steven with a raised hand and then

anyone else who wants to raise their
hand or a question in the chat

i'd like to bring in a bit more on that
aesthetics and embodiment i put a couple

of notes actually in the in the in the
youtube chat as well just uh with the

link i i don't know if you've heard of

the william morris and philip webb

red house in bexley heath ken um it

talks a lot about the crafts because

they actually

um they had this whole thing in the

1800s where they were

wanting to redo

the crafts in the in the house and

actually had little holes in the ceiling

so they could redesign the ceilings

in a craft way for different events

so there's this push against

industrialization so i'm wondering when

you mention about

the better aesthetics by putting the

corners i mean

in some ways you you you gain what

could be seen at one level of aesthetic
but there again with a craft and seeing
some of the variation
the the the reach
and i wonder how you feel about this the
reach of the embodied
inference
of the person making it
is is lessened so the ability to i
wonder what your thoughts are about the
ability to infer from the other
direction
what the process of creation was and how
that aesthetic creates more soul how
that is in a way what the soul of
objects are
um
so if
let me
[Music]
try to sum up what
my understanding is of what you what you
were saying
so my understanding of what you're
saying correct me if i'm wrong is that
if something is
crafted
then it has more
um
inherent
value at least to the person doing the
who crafted it so if somebody carves a
carves a sculpture
then
that has
more inherent value to them
than if they
downloaded a model of a sculpture and 3d

printed it

is that

yeah there's and and on top of that the
ability for their embodiment of creation
their process of creation or their
active inference or an active inference
process of creation extends further out
into that distributed network because
someone who's further removed from that
person could see that sculpture
and they could
get closer to being them than being
closer to being a cdc
routing machine operator

[Music]

well i don't know because if i see it if
i see a sculpture
then
i can appreciate it hopefully
but i'm far from
being able to
do it myself
and uh so let me give you a more mundane
example
that
in many parts of the world walls used to
have walls and ceilings used to have
wet plaster applied to them to make them
smooth
and
so
plastering a ceiling
getting the plaster on the ceiling the
wet plaster
on the ceiling and it's smooth
it's a very difficult thing to do
and uh so people will do apprenticeships
as plasterers and so on

and
so i can see that i can see oh yeah that
looks and maybe i could try to do it
myself as well and all the plaster would
end up on the floor and on me
and then i'd realize that oh this is
really difficult
but
it doesn't get me any closer to being a
plasterer with the skill to to get the
plaster on the ceiling
and if all dwellings depend upon the
construction of all dwellings depends
upon such processes
then
billions of people are going to remain
main in
uh
bad
dwellings or no dwellings so
that's the so if i understood you
correctly i don't think it makes me but
if i see somebody
putting up um
some boards on a ceiling that i've
designed to give a
a functionally um
adequate and aesthetically adequate
ceiling
and i see them do that then i can do it
cool thanks just one just just one just
one follow-up just very very quickly i
think i suppose i could also i'm not
necessarily disagreeing with what you're
saying at all i think but i would also
just
in terms of this distributed
um like if you go into a medieval house

and you see
a more rough and ready plastering you
may not know how to do it but
you could imagine the process of someone
in bodies that did it even if you
couldn't do it yourself and so i was
wondering about that
those kind of those removals as things
get industrialized
which may be you know
maybe that's why sometimes we add
texture back in
and sometimes more expensive stuff that
we buy in the shops actually has texture
put back in that can be actually
expensive to produce
it reminds me of graffiti like when you
see graffiti you can imagine an
affordance that would lead to you having
done that art
versus like a you know an apple
phone it's delivered like in the conch
shell
with smooth and you know it's sort of
the extreme case uh where at the very
least it's more cognitive to understand
how it was built and it definitely makes
sense and things that are well worn they
uh it doesn't make them like better or
more valuable just makes them
more able to play a part of a story in
someone's life like putting up a kid's
drawing
it becomes about the person and latching
on to a broader network of connections
versus the very
parsimonious and ruthless efficiency of
an industrial design process that kind

of seeks to
um remove
the human touch and like you know if it
was a craft beer it has a little dirt
its character
it's a mass-produced one has a little
dirt it's dirty
so we tolerate um different types of
imprecision
when we know that we're dealing with
like our neighbor versus when we're
expecting some
international
shipment of uh food
um so scott with a question and then
looks like you want to share something
else stephen
you know it's a it's interesting it's
kind of the disgust reaction is invoked
there too and that idea of um
that rejection and
anyway the um
that reminded me of that last bit was in
uh there's a black box of ai that we
talk about now well in way we have a
black box of industrialization
right the thing was made but we don't
know how
and it used to be that if you look at
this if you look at a shoe you can say i
see the stitches and i see there's a cow
over there i can see how you get leather
and i get it then you look at a nike
shoe and you're like i have no idea
where that came from how it was made if
it was glued what was put together right
it's just so in a way
we have that black box now me when marx

was talking about alienation
and industrialization maybe part of our
alienation is not
knowing our connection to craft
one of the things i talk to people about
is in europe they have these rights
called drop morale they're moral rights
it's a copyright right overlay
so if i sell you a painting in europe
you can't destroy it i still have rights
as an artist moral rights in the art is
a form of expression i think it has a
hegelian and kantian background but
let's put that aside and then
in the united states if i sell you a
painting you do anything you want to it
we have a lock and utilitarian notion of
these
expression and rights and individual
rights so it feels like there's
indifference here philosophically in
terms of connection to expressive
content
in the drought morale context you're not
talking about industrial production
you're talking about copyright which
gets towards industrialization of
guild type production right because it's
a work of original authorship
again maybe not directly related but i
just thought it was really interesting
that in a way we're looking at that
black box question again and humans
being alienated from production
and i wonder if there's some aspects in
in the work that we're seeing here of
that
de-alienating production i wouldn't

comment on that nice question scott

thank you

can you know can you see a diagram it's
sort of got a slopey line and some black
boxes all right

so

it says here

increasing geographical and social
distribution of production

so

d growth now d growth centralized
industrial production this is happening
with premature deindustrialization
in in many countries in the world they
had some industrialization and now it's
it's going

and we see that

what's coming on is there's a
distributed industrial distributed
artisanal and distributed
diy that's growing

and

what but probably what's still going to
come from centralized industrial
production is materials form materials
and then components
and then

local goods is becoming more distributed
then distributed infrastructure
and then down here

[Music]

this so

industrial production

so this requires massive capital
investment in fixed industrial
manufacturing assets

then you get industrial mass production
and this is all based upon that you get

higher productivity from that massive
capital investment

but can this produce higher wages more
goods purchased is there still scope for
that

will this lead to increased
a production employment evidence
suggests not

so this is non-industrial production
for one of a better term

so with this there's minimal capital
investment in flexible

non-industrial production assets
more industrial production then there's
higher productivity

and that's

coming

and all the options here is you can have
higher value artisanal production

so if you

you know if shoes are handmade
to some extent

uh

as they are the company doing that those
in ethiopia

[Music]

that can lead to higher wages

and that's more goods and infrastructure
made with personal commitment

leading to increased paid and personal
production employment

and then down

lower price diy production lower
expenditure

so with

presumption there are you could for
example

do produce something with some degree

of craft in it
and that could produce a higher
sales price higher wages and then at the
same time
your costs can be reduced so if you
fabricate your own solar panels and you
install them yourselves
and suppose you can get all your
electricity from that well then your
energy costs go down so i'm seeing this
has been um is that diagram
very visible in your
yep i made it large
it's what you see on jitsi is not what
the live stream is
okay
all right so so that diagram i think is
addressing
um
both sides of this that on the one hand
you're
reducing the this lower price diy
so there the idea is to
eliminate all redundant variation
[Music]
to drive down costs of
essential goods
as low as possible so as many people can
access them as possible so they can um
fabricate
and install themselves
and
the higher value artisanal production
this
this is this using uh
automated engineering this mass
imagineering type of thing
pretty so cool reminds me of doing more

with less and also bucky um
so maybe we can just each have a final
thought or a final area that we want to
explore a little bit
next week
and then otherwise this has been an
awesome
discussion
so
um
you just think for a few seconds for
what would what would be our what was
something that will stick with us from
today or where do we want to take it in
our dot two so um scott first then dean
i really like that
concluding thought on that really feels
like that
conversion of labor and attention and
skills
and the into embodied our embodied world
and that feels like that dynamic is
going to be very interesting to continue
to explore in that last diagram there
those pathways and the complementary
pathways are fascinating and thank you
for fantastic discussion
thank you scott dean
yeah i just wanted to i have to read
this because i was listening and
thinking about what i wanted to close
with um my takeaway today
besides the fact that i've got a new
reference with embedded jet boots that
i'm gonna have to go and and try to hold
myself down with um
is that perhaps life is an
apprenticeship

uh in wayfinding across scales um
so one of the things that i think that
we're trying to do in these live streams
is
popularize active inference we're trying
to drive down the costs per se
of that entry
barrier
and so
these kind of interaction processes like
we're having today in this point one
i think are giving us more of a sense of
how that
cost
factor can be addressed
um
beyond a room full of academics which i
think we all agree is kind of a small
niche
and i think that's something that we're
all trying to make available to more
people so
that last reference that you made
scott was actually pretty
pretty helpful
um i think one of the things that we try
to do in terms of an interactionist
uh forward approach is in the point
zeros we deconstruct
and the point ones we reconstruct and
then in the point twos which is next
tuesday which i'm unfortunately going to
be
on a
i'm going to be away from but that's
where the new or next superordinates
sort of emerge and so that i think the
way that we're trying to set this up

whether it's going to
i don't know when when it's going to
sort of
take off like when the jet boots will
really be fired up but i think we're
putting things in place for that
to happen so i think on the one hand i
can i can kind of laugh about the fact
that i'm going to have to anchor myself
but on the other half vladimir says
anytime we can get creative with this
stuff
it's it's worthy it's it's enthusiasm
worthy so thank you for sharing with us
today i really appreciate it
thank you dean
um stephen and then blue
thank you again yeah i really enjoyed it
today is a great great call and um
i'm going to take away the idea of
situated entropy i think that's really
useful
and has a bit more a lot of legs to it
and i'm really interested in more about
talking about pragmatic approaches to
using the variation expected free energy
i think that goes beyond what was often
maybe
pragmatism which we've had here as a
kind of philosophical thing but maybe
pragmatics in terms of design and
application so thank you brilliant
more than just my closing thought is uh
i'm really fascinated by your experience
and theorizing and other publications on
this
the past and present and future of that
continuum or multi-dimensional space

between craft and artisanship the
boutique the bespoke
everything on that principal component
side and then
industrialization and
have we hit peak something you know have
we seen the biggest solid state drives
of all time in 2019 that was peak solid
state or
what will be peak what and what's been
going to be what are we at the bottom of
what are we near the top of
and some of the the roads and the paths
that you sketched are pretty cool so
steven fox if you have any final
comments
uh scott could you email me a link to
the organization that you you were
mentioning
i will uh email um
i'll email steven's email address to
scott
okay thank you very much
uh yeah well thank you thank you all
very much for taking the trouble to read
my
my paper
because it
i was plowing a
lonely photo when i was doing it
lou with the last word
thanks steven and you know you gave us
so many additional references to read
that that's really what i'm looking
forward to doing between now and next
week is digging into some of this
additional literature that i've been
made aware of

cool so till next week we'll read the
papers think of questions invite any
colleagues or anyone else so all
backgrounds are welcome to join so we'll
see you all next week

bye

thanks bye bye

you